Stochastics and Machine Learning **ETH Zürich**
Spring 2024 **Cheridito, Cotrini and Streich**

Problem Set 3: Random Variables and their Distributions

(March 8 – 21, 18:00)

Problem 1 (Remarkable Distributions 1)

a) Compute $I = \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}}\,dx$.

Hint: Write $I^2 = \left(\int_{-\infty}^{\infty} e^{-\frac{x^2}{2}}\,dx\right)\left(\int_{-\infty}^{\infty} e^{-\frac{y^2}{2}}\,dy\right)$, apply Fubini's Theorem and use polar coordinates.

b) Compute the variance of a standard normal random variable.

c) Compute the expectation of an exponential random variable with parameter λ.

d) Compute the variance of an exponential random variable with parameter λ.

Problem 2 (Remarkable Distributions 2)

a) Let $X \sim$ Pois(λ) and $Y \sim$ Pois(μ) be two independent Poisson random variables. Show that $X + Y \sim$ Pois$(\lambda + \mu)$. **Hint:** Derive and identify $\mathbb{P}(X + Y = k)$, for $k \in \mathbb{N}$.

b) Let $X \sim$ Exp(λ) and $Y \sim$ Exp(μ) be two independent exponential random variables. Show that $\min(X, Y) \sim$ Exp$(\lambda + \mu)$. **Hint:** It is enough to show that the distribution function of $\min(X, Y)$ coincides with the distribution function of an exponential with parameter $\lambda + \mu$.

Problem 3 (Rayleigh Distribution)

The velocity X of a particle of mass m is modeled by the so-called *Rayleigh distribution*, i.e., X has the density

$$f(x) = \begin{cases} xe^{-x^2/2}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

a) Show that f is a density function.

b) Calculate the cumulative distribution function F_X of X.

c) Calculate the probability that the particle has a velocity between 2 and 5.

d) Calculate the cumulative distribution function of the kinetic energy $Y = \frac{1}{2}mX^2$.

e) What is the name of the distribution of Y?

f) Compute $\mathbb{E}[Y]$.

Problem 4 (Construction Sites)

In the city of Zurich, there are famously many construction sites. The duration X of the work at a construction site ranges from 0 to 20 weeks. The density $f(x)$ has the following form:

$$f(x) = \begin{cases} c - \frac{c}{20}x & \text{if } x \in [0, 20] \\ 0 & \text{otherwise.} \end{cases}$$

a) What is c and why?

b) Calculate the cumulative distribution function and sketch it.

c) Determine, using the cumulative distribution function, the probability that the construction time X *(i)* does not exceed 5 weeks, *(ii)* lies between 5 and 10 weeks.

d) Calculate the expectation, the median, and the standard deviation of the duration X.

e) What duration is exceeded with only a 10% probability?

f) Let $K = 40,000 \cdot \sqrt{X}$ represent the amount in Swiss Francs that the work at a construction site costs. What is the probability that the work at a construction site costs at most 120,000 Swiss Francs?

Problem 5 (Insurance Model)

A commonly used model in insurance to cover large losses is a so-called "Excess-of-Loss" contract. Against payment of a premium, the reinsurance company undertakes to cover any losses that exceed a certain level of x_0 CHF.

To determine the amount of the premium, the company examines the large losses (losses $> x_0$) of the past year. Such large losses can be modeled by a *Pareto-distributed* random variable X, where for a parameter $\alpha > 0$, the cumulative distribution function of X is given by

$$F_X(x; x_0, \alpha) = \begin{cases} 1 - \left(\frac{x}{x_0}\right)^{-\alpha}, & x \geq x_0, \\ 0, & \text{otherwise.} \end{cases}$$

1. What is the average cost of a single large loss?
2. The net annual premium is calculated by $P_{\text{net}} = \mathbb{E}[X]\mathbb{E}[N]$, where $N \sim \text{Pois}(\lambda)$ models the (random) number of large losses per year. How large is the premium P_{net} for $\alpha = 2,\ x_0 = 2 \cdot 10^6$ CHF, and $\lambda = 3$?
3. Why is $\alpha > 1$ typically assumed in the above model?

Problem 6 (Multiple Choice)

Which of the statements below is correct? Justify your answer.

a) $\mathbb{P}[X > t + s \mid X > s] = \mathbb{P}[X > t]$ for all $t, s \geq 0$, if

1. $X \sim \text{Unif}(a, b)$.
2. $X \sim \text{Pois}(\lambda)$.
3. $X \sim \text{Exp}(\lambda)$.

b) Let $\Phi(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{t} e^{-\frac{1}{2}x^2} dx$. What is $\Phi(-t)$?

1. $\Phi(-t) = -\Phi(t)$.
2. $\Phi(-t) = 1 - \Phi(t)$.
3. $\Phi(-t) = \Phi(t)$.

c) Let $X \sim \mathcal{N}(0, 1)$ and Y be a random variable such that $X + Y \sim \mathcal{N}(1, 6)$. Calculate $\mathbb{E}[Y]$.

1. $\mathbb{E}[Y] = 0$.
2. $\mathbb{E}[Y] = 1$.
3. $\mathbb{E}[Y] = 2$.

d) The density of a random variable X is given by

$$f_X(x) = \begin{cases} c+x & \text{if } -\frac{c}{2} \le x \le 0, \\ c-x & \text{if } 0 \le x \le \frac{c}{2}, \\ 0 & \text{otherwise.} \end{cases}$$

Determine the constant c.

1. $c = \frac{2}{\sqrt{3}}$.
2. $c = \frac{1}{\sqrt{3}}$.
3. $c = \frac{3}{\sqrt{2}}$.

e) Consider the distribution function $F_X(t)$ for the random variable X from d). Then:

1. $F_X(t) = \frac{1}{2}t^2 + \frac{2}{\sqrt{3}}t + \frac{1}{2}$ for $t \le 0$.
2. $F_X(t) = -\frac{1}{2}t^2 + \frac{2}{\sqrt{3}}t + \frac{1}{2}$ for $t \ge 0$.
3. Neither 1. nor 2. is correct.

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use this

a) $\mathbb{P}(X+Y=k) = \sum_{j=0}^{k} \mathbb{P}(X+Y=k \mid Y=j)\,\mathbb{P}(Y=j)$

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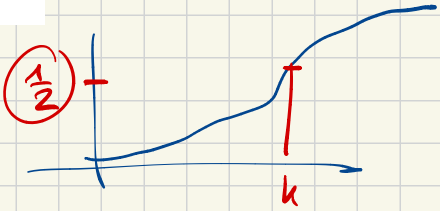
a) remember: $\int_{-\infty}^{\infty} f(x)\, dx = 1$



d) Median: $\underbrace{F(x = \text{median})} = \frac{1}{2}$

e)

f) we search for $P(K \leq 120'000)$

↳ plug in X

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3. Why is $\alpha > 1$ typically assumed in the above model?

1) $f_X(x) = F_X'(x)$

2. if $N \sim \text{Pois}(\lambda) \Rightarrow E(N) =$

3. counterargument: $a \geq 1 \Rightarrow$

Problem 5

Very exam-like (maybe on the easier side)